

Deep Learning and Heterogeneous Agent Macroeconomics

Reading List

Penn State Mini-Course · September 2026

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Before the Mini-Course (Suggested Pre-Reading & Pre-Coding)

The lectures will be much easier to follow — and the discussion much more fun — if you have had a look at these two items beforehand. If your time is limited, the coding item is the one that pays off most.

- **Reading — heterogeneous-agent models.** Dirk Krueger, *An Introduction to Macroeconomics with Household Heterogeneity* (lecture notes), 2026. [\[link\]](#)
Chapter 6, “The SIM in General Equilibrium,” is the most directly relevant; Chapter 4, “The SIM in Partial Equilibrium,” is a useful background reference if the material is new to you.
- **Coding — neural networks for economic models.** Simon Scheidegger, tutorials on using neural networks to solve simple Brock–Mirman models (Jupyter notebooks).
If you can, read and run both notebooks: Notebook 1 — [deterministic Brock–Mirman](#), and Notebook 2 — [stochastic Brock–Mirman](#). New to Python? See the [Python refresher](#) here.

Lecture 1 — Deep Learning for Solving Heterogeneous Agents Models (DeepHAM)

- Han, Yang & E (2026). *DeepHAM: A Global Solution Method for Heterogeneous Agent Models with Aggregate Shocks*. Quantitative Economics 17(2), 297–341. Code: github.com/frankhan91/DeepHAM. [\[link\]](#)
Neural networks approximate the value and policy functions, and the cross-sectional distribution is represented by a small set of *learned* generalized moments; the objective is optimized directly along simulated paths. Focus on the moment representation, the simulation-based training objective, and the constrained-efficiency application.

Lecture 2 — Deep Learning for Continuous Time Models and Structural Estimation (DeepSAM)

- Payne, Rebei & Yang (2026). *Deep Learning for Search and Matching Models*. Conditionally accepted, Econometrica. [\[link\]](#)
Characterizes general equilibrium of search and matching models with aggregate shocks and two-sided heterogeneity as a high-dimensional PDE with the distribution as a state variable, then solves it globally by deep learning and estimates parameters via simulated method of moments. Applications to labor search: distributional feedback and amplification, and sorting over the business cycle.

Background for the Whole Course

Heterogeneous-agent models

- Aiyagari (1994). *Uninsured Idiosyncratic Risk and Aggregate Saving*. Quarterly Journal of Economics 109(3), 659–684. [\[link\]](#)
The foundational discrete-time, incomplete-markets heterogeneous-agent model.
- Krusell & Smith (1998). *Income and Wealth Heterogeneity in the Macroeconomy*. Journal of Political Economy 106(5), 867–896. [\[link\]](#)
The classic heterogeneous-agent model with aggregate risk. Its curse of dimensionality is exactly what Lecture 1 is designed to overcome, and it is the lead application there.

Deep learning for economic models

- Fernández-Villaverde (2026). *Deep Learning for Solving Economic Models*. Journal of Economic Literature, forthcoming. [\[link\]](#)
A broad survey of how deep learning is used to solve dynamic economic models, with code on the [companion page](#).
- Maliar, Maliar & Winant (2021). *Deep Learning for Solving Dynamic Economic Models*. Journal of Monetary Economics 122, 76–101. Code: github.com/marcmalier/deep-learning-euler-method-krusell-smith. [\[link\]](#)
The “all-in-one expectation operator” formulation: train a neural network on simulated paths to satisfy the model’s equilibrium conditions.
- Azinovic, Gaegauf & Scheidegger (2022). *Deep Equilibrium Nets*. International Economic Review 63(4), 1471–1525. Code: github.com/sischei/DeepEquilibriumNets. [\[link\]](#)
Neural networks trained to satisfy equilibrium conditions along simulated paths, with a wide range of applications.
- Goodfellow, Bengio & Courville (2016). *Deep Learning*. MIT Press (full text free online). [\[link\]](#)
The canonical deep-learning textbook.
- Gu, Laurière, Merkel & Payne (2024). *Global Solutions to Master Equations for Continuous Time Heterogeneous Agent Macroeconomic Models*. arXiv:2406.13726 (forthcoming, Journal of Political Economy: Macroeconomics). [\[link\]](#)
Introduces Economic-Model-Informed Neural Networks (EMINNs) to compute global solutions of master equations, with the continuous-time Krusell–Smith model as the lead application.
- Kargar, Passadore, Silva & Yang (2026). *Liquidity and Risk in OTC Markets: A Theory of Asset Pricing and Portfolio Flows*. Working paper. [\[link\]](#)
An OTC application of the search-and-matching machinery in Lecture 2: asset pricing and portfolio flows in over-the-counter markets with liquidity and risk.